Handling instructions
Confidential

Industry-University Joint Research

Study on the Demonstration of the Effectiveness of **eco-SPRAY®** on Vessels

First Report

Impact on ship engines and evaluation - Verification

(Source: Japan Family Support Co.
(Hiroshima Merchant Marine Technical College
(Production) EiShin Co.

March 1, 2024

Research Background

◆Goals and measures to address global climate change and decarbonize the shipping industry

With the growing momentum for global decarbonization, Japan has announced its policy to achieve virtually zero GHG emissions by 2050. At the Climate Change Summit to be held in April2021, the government announced its policy to reduce GHG emissions to 46% below the 2013 level, and accelerated its efforts.

Establish a system to set emission quotas for companies and individual facilities to reduce emissions efficiently and impose reduction obligations (excesses and deficiencies can be traded on the market).

Companies can choose to reduce emissions through self-help efforts or purchase allowances.

In the EU, a review bill to extend the Emissions Trading Scheme (EU-ETS) to the shipping sector will be submitted in 2021/7.

Growing global momentum for decarbonization drives decarbonization in the shipping industry

Regulations were studied and introduced in ocean shipping under the GHG reduction targets set at the initiative of IMO. After 2021/4, each country is recommended to raise the target.

Domestic shipping is also expected to review its reduction targets, strengthen efforts to switch fuels, and increase environmental investments.

CO_2 emissions from international shipping (ocean shipping) are equivalent to 67% of Japan's total CO_2 emissions, and as for Japan's coastal shipping, it accounts for 4.9% of Japan's transportation sector CO_2 emissions (about 1% of Japan's total). (Reference: Automobile 86.2%, Airline 5.0%, Railroad 3.9%).

For many years, ships have used petroleum-derived fuels such as heavy oil and light oil, which emit substances that are known to cause global warming and air pollution, and in recent years the marine transportation industry has been obliged to consider the environmental burden. The International Maritime Organization (IMO), a specialized agency of the United Nations, announced in 2018 that it would reduce greenhouse gas emissions from international shipping. Emissions to 2050 The government has announced a goal of halving GHG emissions by 2025 and achieving zero GHG emissions as soon as possible this century. Domestically, the Ministry of Land, Infrastructure, Transport and Tourism has set a goal of having zero-emission ships in service by 2028. Therefore, it is very important for shipping to find fuels with virtually zero GHG emissions, but at present, solar and wind power systems installed on ships cannot generate enough energy to power large vessels. In addition, although the policy for existing vessels is to switch to more environmentally friendly fuels and to promote slow steaming, there are many issues of concern for shipping companies and shipowners, such as increased costs and lead time problems.

From "Current **CO_2** Emissions from Coastal Shipping and the Surrounding Environment," Ministry of Land, Infrastructure, Transport and Tourism

Potential of ecoSPRAY's Shipboard Utilization Business

◆Background of the research

The shipping industry is currently facing a difficult environment, but on the other hand, trade volume and cargo movements are expected to increase, given the current situation in which the world's population continues to grow. In other words, the shipping industry can be viewed as a growth industry, one that certainly has great potential for the development of environment-related businesses. (hereinafter referred to as " EiShin"), Medi-Science Espoir Corporation, and Japan Family Support Corporation (hereinafter referred to as "the three companies' joint research group") decided to conduct a joint research on the effects of eco-spray (an eco-friendly commercial product developed to reduce automobile exhaust gas and improve fuel efficiency) on marine vessels, with the cooperation of Hiroshima Maritime Institute (hereinafter referred to as "Hiroshima Maritime College"). (hereafter referred to as "Hiroshima Merchant Marine") to conduct a joint research on the effects of eco-spray on the main engine of ships. The joint research on "Research on Demonstration of Effectiveness of eco SPRAY on Ships" was conducted at the school from September2022, and the results of the research were reported in December 2023.

◆What does eco-SPRAY do?

Action of eco-SPRAY - Effects on automobile engines

1. Increased explosive power in the cylinder
2. the surfactant activity of hydroxyl ions provides a cleansing effect.
3. due to the composition of seaweed, when metals of divalent or more are bound, they become gelatinized and insoluble in water and are discharged out of the muffler. This removes unnecessary substances from inside the engine, making the engine work harder and improving combustion efficiency.

Continued use of automobiles causes incomplete combustion due to deterioration of engine performance and combustion debris stuck inside the engine, resulting in emission of toxic gases. Continued use of eco-SPRAY improves combustion efficiency through the action of the above components, reducing harmful emissions, and increasing power and fuel economy by making the engine run better.

Assumption:
The same thing should happen with ship engines. Joint research begins to demonstrate impact and effectiveness on ship engines.

Similarities and differences between automobiles and ships

(1) Engine powered (1)
heavy fuel oil for opera

etc. (4) There is a filter
throughthe air intake

andbulk carriers (5) Exhau

What is eco-SPRAY (overview)

eco-SPRAY is an innovative product developed by EiShin Co., Ltd. to improve vehicle fuel efficiency and reduce exhaust emissions. Instead of adding eco supray to engine oil or gasoline, simply spraying eco-supray on the air filter of an automobile reduces exhaust emissions and improves fuel efficiency. The effect lasts approximately 5000 km under normal use. The **ingredients are 100% naturally derived**, so they are gentle to people and the environment and can be used safely and securely.

◆Function of ingredients

1.	One of the ores in the ingredients passes through the air filter and is sucked into the cylinder, increasing the explosive power. In addition, hydroxyl ions generate a surfactant effect to clean the inside of the engine.
2.	Exhaust gas cleaning effect due to Hinokitiol, etc., which acts as a natural antiseptic.
3.	The seaweed component makes the ore adhere to the filter. When metals of divalent or higher are bound, they become gelatinized and insoluble in water and are discharged from the muffler. Seaweed ingredients reinforce metal bonds.
4.	2 and 3 remove unnecessary substances from inside the engine, making the engine work better and improving combustion efficiency.
5.	Oxygenated WOX® water: Dissolved oxygenated water developed with our own special technology. Prevents separation of natural components and improves filter penetration for complete combustion.
6.	Strongly stabilized silver ions "HT Silver®": Ensures stability of eco-spray quality. It does not change color or lose its effect due to light, heat, acid, or alkali, does not rust, does not damage plastics, and does not affect paint staining.

◆The World recognized eco-Spray technology

The eco-SPRAY technology was **patented in Japan in 2018 and in the United States in 2021**. This product has been recognized by UNIDO (United Nations Industrial Development Organization) Tokyo Office for its technology and is registered in the United Nations "Environmental Technology Database". In addition, the SDG 's logo (with a designated sentence) can be used by introducing this product, so it can be used as a corporate response to the SDG's and CSR. As a result, the company has expanded its business to many countries, including the U.S., China, India, South Korea, and the Philippines, and is now promoting sales in Japan as well.

Inauguration of industry-academia joint research

◆ Joint Research Group – Industry & University

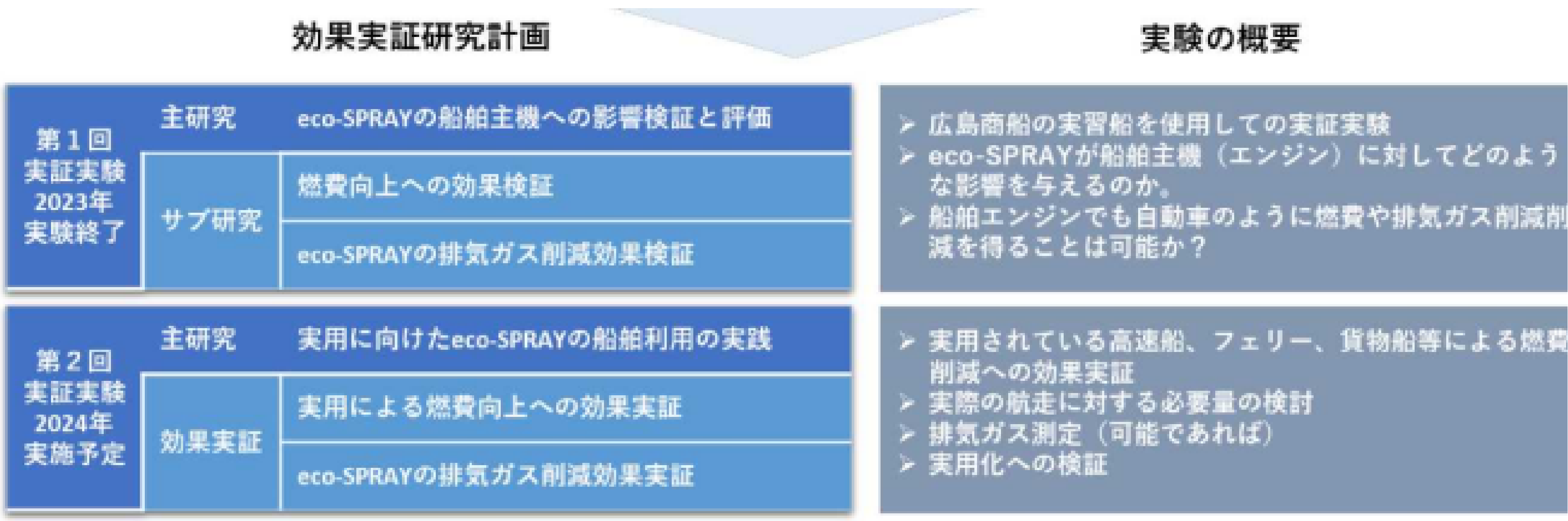
In order to verify whether eco-SPRAY is effective on ships, a joint research group of three companies was formed with EiShin Co. Ltd. and Mediscience Espowa Co., Ltd. to conduct and coordinate the research. A joint research agreement was concluded with Hiroshima Merchant Marine to provide a test ship and cooperation in conducting experiments, and full-scale joint research was initiated using the school's own vessels (training ships, etc.). In the final stage of the second demonstration experiment, an engine disassembly by an engine manufacturer (Komatsu) was incorporated, and the effect of eco-Spray on the engine interior was examined.

Joint Research Group – Industry and University

Japan Family Support Co.	Research Implementation - Coordination Contact
㈱ Eishin	eco-SPRAY Raw Material Compounding - Manufacturing
Medience-Espore Corp.	eco-SPRAY Raw Material Compounding - Manufacturing

Assistant Professor Iwakiri, Department of Merchant Marine,	Experimental Site Supervision - Site Supervision Experimental -
Hiroshima National College of Ishida Shipbuilding Co. Maritime Technology, etc.	Verification Construction Execution

Refer to the HP of Hiroshima Maritime College of Technology
[Collaboration | Research activities | Hirotori College of Maritime Technology](hiroshima-cmt.ac.jp)
hiroshima-cmt.ac.jp



Impact on ship's main engine and evaluation

In this study, we conducted an on-board sailing experiment with the aim of applying eco-SPRAY to ships, and in the final stage, we opened the main engine and confirmed the condition inside the cylinder using eco-SPRAY . With the cooperation of Ishida Shipbuilding Co., Ltd. the trainee vessel was put into dock and an open inspection by the main engine manufacturer (Komatsu agent: Kyowa Industries Co., Ltd.) was practiced. As a result, the company received a high rating of **no abnormality observed in the intake and exhaust systems**. This confirms the safety of eco-SPRAY for ship's main engines. The cleaning effect on internal combustion engines was also evident, indicating its effectiveness for marine use. (For the results of this research, please refer to the report by Assistant Professor Iwakiri (Hiroshima Merchant Marine) of the joint research group (Appendix 2).

Port plane

cylinder head Starboard plane cylinder head

The condition inside the combustion chamber is good. Changes were observed in the amount of carbon deposits in the combustion chamber, and the main engine using eco SPRAY® had less carbon deposits.

Improved combustion efficiency

Reduction of hazardous exhaust emissions

Approved Cleaning Efficiency

Third-party verification and evaluation

Ishida Shipbuilding Co.

A long-established shipbuilding company that celebrated its 105th anniversary in May2024. In addition to private companies, the company also handles the docking of ships for the Maritime Self Defense Force, the Ministry of Land, Infrastructure, Transport and Tourism, the National Safety Agency, other government agencies and private companies.

Kyowa Industries Co.

A company that sells and repairs Komatsu and Kobelco construction equipment, and inspects, repairs, and overhauls marine engines, generators, gas turbines, and other equipment.

ımage was observed to marine engines by eco-SPRAY, and the cleaning was confirmed for marine engines as well as for automobiles. From this, it e inferred that the improvement of combustion efficiency and reduction of ıst gas emissions can be achieved with eco-SPRAY. In fact, this iment also showed an improvement in fuel efficiency, and the clean ıst emissions were visually confirmed. (P.8 Overview of Joint Research nents" thereafter).

'est was completed without any problems regarding the effect on the ıe. The effectiveness of eco-Spray in marine engines was confirmed.

Future Issues and Plans

As a result of the joint research, **it was determined that the use of eco-SPRAY in ship engines was not problematic, and its effectiveness in cleaning the inside of engines was also demonstrated.**

The impact on engines was the subject of this study, but fuel consumption reductions were also measured in the process. The status of exhaust emissions is also checked, albeit visually. Please refer to the "Summary of Collaborative Research Experiments" starting on the next page. Since it has been proven that there is no impact on ship engines, as the second step, verification will be conducted on ships actually used in the project in order to effectively improve fuel efficiency and reduce exhaust gas emissions. Although there are some issues to be addressed, this research has already proven to be effective for ship's main engines, so the potential for practical application is high.

◆Future issues

How to apply eco-SPRAY This time, gauze coated with eco-SPRAY was wrapped around the engine filter to measure fuel consumption. Should we apply directly to the filter next time? The amount and frequency of application (once every7,000 km to 10,000 km for diesel vehicles) for marine engines of various sizes

Fuel Economy Measurement Methods This time, the experiment utilized two units of one ship's engine. Since both engines were in the same engine room and the filter with eco-spray

CO HC NOx CO_2 monoxide Hydrocarbon Nitrogen

Carbon

Carbon

oxides

dioxide

and the filter without eco-spray were in close proximity, it is possible that some of the components were dispersed to the side without eco-spray. How will the next practical

30% I 23% I 15% I 0,43% I

experiment be measured? Experiment with two vessels sailing the same distance for the same amount of time? We need to find companies that are willing to cooperate with us. Since ships sail on the sea and are greatly affected by ocean waves, currents, and winds, it is impossible to measure fuel efficiency improvement in terms of mileage, as is the case with automobiles. Should the time to run be used as the criterion? **Exhaust gas measurement method** The exhaust ports on vessels are either above the ceiling or discharged from the rear with seawater, neither of which can be easily measured as in a car. Incidentally, vehicle exhaust emissions have been measured using gasoline by JATA [Japan Automobile Transport Technology Association]. (See p. 14)

◆Future plans

In the future, we will work with shipping companies that are willing to cooperate with our efforts to find effective ways to use this technology on vessels. The joint research group for shipboard use of eco-SPRAY will continue its activities with the mission of reducing environmental impact and contributing to global warming and the creation of a sustainable society in line with the goals of **the SDGs**.

Photo: "Hikari" from the website of Tanaka Shipbuilding Co.

Commencement of joint research

◆**Outline of joint research: Significance and necessity of conducting this joint research**

The fuel additive " eco-SPRAY " was developed by EiShin Corporation, a co-researcher of this study, and its technology has been patented in Japan and the U.S., and is recognized in the UNIDO database as an environmental technology that can be deployed globally. This improves combustion efficiency by stimulating air activation and removing static electricity and carbon sludge from the engine, thereby reducing the amount of toxic gases emitted from automobile engines (CO2, Co, HC, NOx). In addition, since they are composed entirely of naturally derived ingredients, they are gentle to the environment, human body, animals, and plants. The purpose of this study was to verify whether the " eco-SPRAY " developed for automobile engines could be used in marine engines to achieve the same effect. We will conduct experiments using actual marine engines in a marine environment that differs from that of land, to verify the degree to which combustion efficiency can be improved and harmful emissions reduced, and at the same time, to demonstrate the safety of marine engines. If the effects of this research on ships are proven, it will help the shipping industry to reduce exhaust emissions and fuel consumption, which in turn will help to address the global warming problem and contribute to the " SDGs" proposed by the United Nations, which are efforts to achieve a sustainable society. In addition, the reduction of fuel consumption by improving fuel efficiency is expected to save shipping companies money and contribute significantly to CSR, including environmental protection.

*Research summary citation for joint research application to Hiroshima Merchant Marine Technical College by Japan Family Support Co.

◆Start of pre-collaborative research June 2021

Compared to automobiles, ships are more difficult to replace in the event of breakdowns, and the warranty price is higher, so careful research is needed. Although the safety of using eco-suprays in automobiles has been recognized, this is the first time eco-suprays have been used in ships, and it is important to demonstrate some degree of safety before proceeding to full-scale joint research. Hiroshima Merchant Marine kindly agreed to conduct the experiment on its own work vessels, the "Haruna" and "Shiokaze," in preparation for a full-fledged joint research project. The engines of the two test vessels are outboard engines, which discharge all exhaust gases into the water.

Objective.	Targeting marine conversion of eco-SPRAY, confirming applicability to outboard motors
Place of implementation	Hiroshima Merchant Marine Hiroshima Maru Pier and the waters surrounding Osakigamijima
machinery and tools	Training ship "Hiroshima Maru" onboard work boats "Haruna" and "Shiokaze" (open-topped boats) Yamaha outboard motor 8 C 6 NO-L 8 hp
experimental technique	The experiment was performed on the model of the automobile experiment conducted at JATA. 40ml to 45ml of eco-SPRAY was applied to the Haruna and Shiokaze. The power output was set to about 30 % throttle opening for navigation safety purposes, as the vessels would be traveling in actual sea areas.
Number of Experiments	Total 11 times

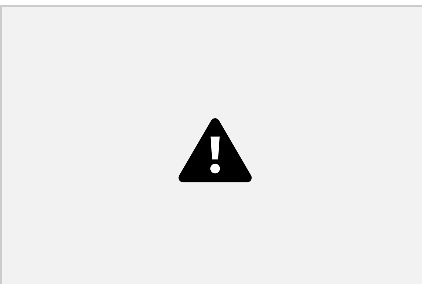
As a result, no engine problems occurred, and the team moved on to the next experiment.

Joint Research: Navigation Experiment No. 1

Since the pre-study confirmed that there was no burden or effect on the engines of the transmission vessel, the team moved on to experiments using a larger vessel as a test ship. The school was asked to conduct all experiments as the vessel was used for educational purposes, and measurements were made in the area around Osaki Kamishima Island in Hiroshima Prefecture. Although exploring the possibility of improving fuel efficiency is an important issue, the primary objective of this experiment was to first investigate and verify the effects on the ship's main engine. The demonstration experiment was conducted twice at different times.

Experiment: Demonstration of effectiveness using the Hiroshima Merchant Marine's small training ship "Hikari" 1st

The small training ship "Hikari" is used. The vessel is a two-engine,two-axle, two-rudder power vessel.



Overall length : 17.40m
Overall width: 04.10m
Registration length : 14.98m
Registration width : 03.80m
Registration depth : 01.62m
Gross tonnage: 16.00t
Main engine output: 281kw x 2 units
Maximum speed: 28.5 knots

eco-Spray

1. Engine type Komatsu 6M108AP-1
2. Uses ¾ bottle of eco-SPRAY (approx. 65 ml) (1 bottle contains 90 ml) 3. Fuel consumption was measured with and without filters, under a range of experimental conditions (weather and sea conditions), and fuel consumption was compared.
4. Experimental Methods The operation time was set to 40 minutes, modeled after JATA's validation method. In actual sea testing, weather conditions have a large impact, so the course is reversed every 20 minutes (to average out the effects of disturbances such as ocean currents and wind and waves).
5. Flow meter used : KEYENCE Clamp-on type flow sensor model FD-H10 (ultrasonic type micro flowmeter)- Accuracy: ±0.15% of rated flowrate

◆Experimental Methods (1) Application of

① ② ③

Since the vessel is a two-engine, two-axle, two-rudder power vessel, it is possible to measure and compare the weather, sea conditions, and other disturbance conditions of the port and starboard engines, as well as the sea area in which they are cruising.

Apply approximately 70% of 90㎖ of eco SPRAY® to 100% cotton gauze thoroughly and soak it

Dry thoroughly.

Create the same situation where the eco SPRAY® component sticks to the filter.

Dry gauze is placed on the turbocharger of the main engine. Since the turbocharger was equipped with a filter to prevent dust and dirt from mixing in, gauze was wrapped over the filter and secured with a band to prevent it from coming off.

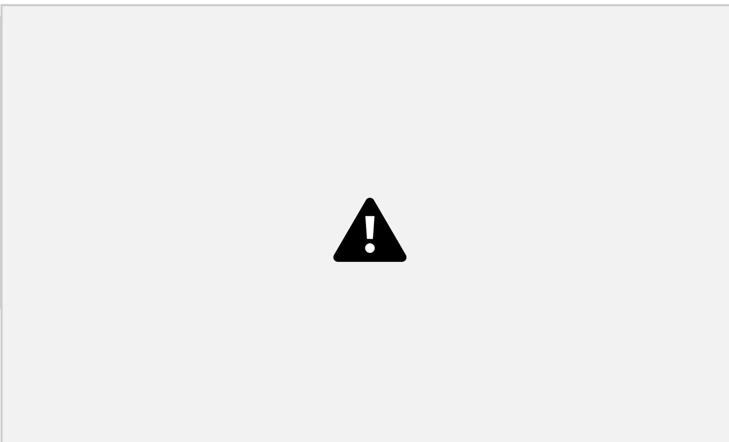
In the case of the automobile, the filter itself is sprayed with eco-SPRAY® to make the ingredients adhere to it, but since the filter is different in form, the experiment was conducted by soaking the filter in gauze and wrapping it around the filter.

Collaborative Research: 1st

on port side

◆Experimental Methods (2) Flowmeter installation

Fuel consumption calculation method Inflow – Discharge = Fuel Consumption



Turbocharger (supercharger) location

Flow meter installed

Install flow meters on the inlet and return sides of the fuel oil lines of the main engine. A total of four units were used for measurements. Fuel consumption is calculated by subtracting the return pipe flow rate from the inlet pipe flow rate.

Engine interior

Main engine type Komatsu 6M108AP-1 eco-spray applied

◆Experimental Methods (3) Measurement by navigation

Demonstration Results

8 times in total

Excerpts from the experimental record (3rd measurement)

power

Experiment	Tuesday, February 1, 2022, 2:29 p.m. - 3:09p
Date weather	cloudiness
wind	West southwest 12.6[m].
direction veering	Pause during needle change.

nim/L

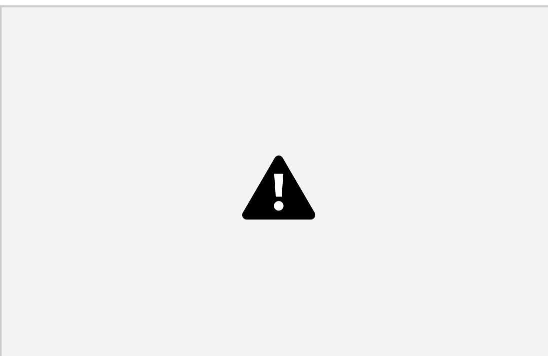
Operating conditions

Main engine speed 2300 rpm

Time: 40 minutes (turnaround time : 20 minutes) From the school's training ship pier to the starting point of the measurement Idling time

Time [min]

and



Seas: Waters around

Osakigamijima, Hiroshima Prefecture

Without eco-Spray With eco-Spray

As shown in the graph on the left, the engines - gauze coated with ecoSPRAY® on the filter - consumed less fuel than the engines without the filter. In this experiment, a 37.5% reduction in fuel consumption was observed.

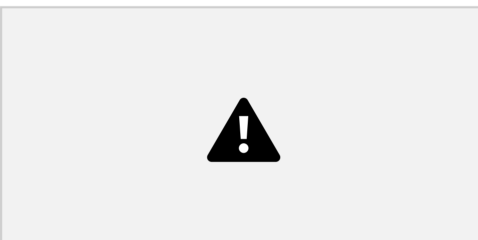
Reduction of 37.5%

Joint Research: Navigation Experiment No. 2

The first demonstration test resulted in a greater-than-expected fuel reduction (37.5%), but both the cruising distance and time were short, so a second demonstration test will be conducted in October2022 as a continuation of the first test. The same training vessel "Hikari" and the same flowmeter were used as in the first trial. A total of 47 cruising experiments were conducted over a relatively long period of time, from October to January of the following year, using almost the same locations for the cruising areas. Later, in August2023, the main engine of the training vessel "Hikari" was opened, and an inspection survey was conducted on the condition of the internal combustion engine.

Experiment: Demonstration of effectiveness using the Hiroshima Merchant Marine Training Vessel "Hikari" 2nd

experimental date	Saturday, October 15, 2022		
test time	going (to)	10:59 - 11:19 a.m.	
	return	11:23 - 11:43 a.m.	
weather	appearing dazzling	wave height	0.1[rn]
wind direction	northeast	atmospheric temperature	29.3rC]
	2.0[m/s]	level of humidity	30(%l
Tide (Direction of flow)	making a tuck	atmospheric pressure	1017 (hPa)



Experiment Date	Thursday, November 3, 2022		
fellow experimenter	going (to)	2: 52 p.m.-3: 12 p.m.	
	distincti	3:21 p.m. - 3: 41 p.m.	

	on		
weather	clear weather	wave height	0.2[m]
wind direction	south-southwest	atmospheric temperature	24.9?C]
wind speed	5.5 [m/s]	level of humidity	341%]
Tide "Direction of flow")	making a tuck	atmospheric pressure	1012.75 [hPa]

A total of 47 experimental measurements.

Although the number of measurements varies from day to day, weather, wind direction, wind speed tides (current direction), wave height, air temperature, temperature, and air pressure are measured. The experimental technique is the same as the first one. The experimental area is almost the same. The amount of eco-SPRAY® applied is the same (approx. 65mℓ). However, while the first experiment was short-term (8 cruises in January2022-February2021), this time it was long-term (47 cruises from October2022 to January2023).

As a result, fuel savings was slightly reduced by 0.32%, possibly due to insufficient application volume, seasonal outside temperatures, and total running time. This point will require further study. For diesel vehicles, the recommendation is to use one bottle every7,000 to 10,000 km. Nevertheless, the total reduction in the first and second rounds was significant, so the outlook for the effect is bright. Although the data on exhaust emissions was not available due to the lack of equipment for measuring exhaust emissions from vessels, it was clear from visual observation that exhaust emissions from eco-sprayed vessels were cleaner, albeit not numerically. (See p. 14)

1st + 2nd demonstration

Total fuel reduction of 19.91%.

Collaboration: Engine open inspection work

In order to verify the **effect on the engine**, which is the most important issue of this joint research, the main engine of the EUT "Hikari" was opened for inspection during the final stage of the experiment. Since ships are not located on land like automobiles, we entered the dock of **Ishida Shipbuilding**, a joint research partner, to open the main engine. Since the engines were manufactured by "Komatsu", we had a representative of **Kyowa Industries Co.** As a result, the internal combustion engine was evaluated to be in good condition and no corrosion or other effects were observed due to the use of eco-SPRAY.

agency operating logs (e.g. factory, agency, etc.)

Trial operation record after

Exhaust manifold open inspection

restoration of the main unit. Engine Division, Kyowa Industry Co.

- Intake manifold open inspection

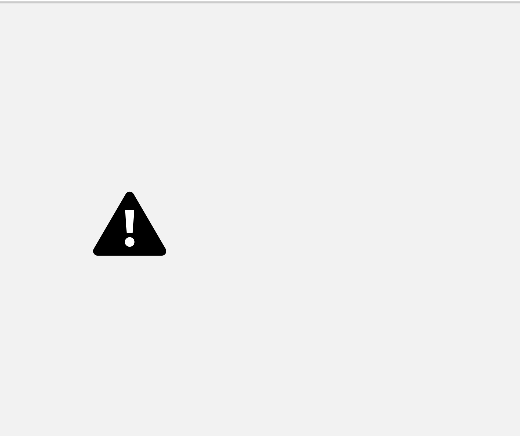
Collaboration: Exhaust Gas

Emissions from ships are emitted from diesel engines, etc., and have been pointed out as an environmental problem. Some of them affect the respiratory tract at high concentrations and can cause acid rain if spread over a wide area. Therefore, the "Regulations for the Prevention of Air Pollution from Ships (Annex VI)" was adopted in the 1997 Protocol. Most of the greenhouse gas emissions from international shipping are CO2, and according to a study by the International Maritime Organization (IMO), emissions in FY2007 were said to be about 870 million tons, which is about 3% of the world's total CO2 emissions, or the equivalent of the emissions of one German country. The global need for maritime logistics is expected to continue to increase in the future, and as a result, CO2 emissions are also expected to tend to increase, making the reduction of CO2 emissions from international shipping an urgent issue. Under these circumstances, the International Maritime Organization (IMO)76th Commission for the Protection of the Marine Environment in 2021 has initiated regulations for CO2 emissions from 2023 for large existing ocean-going vessels around the world. **eco-SPRAY was developed to reduce exhaust emissions and improve combustion by removing deposits from internal combustion engines by inducing complete combustion with the contained components.** Although numerical data could not be obtained in this study, the amount of carbon deposits changed as the internal combustion engine was opened, and it is considered that toxic gases were reduced and CO2 emissions were also reduced due to improved fuel efficiency.

The following data was obtained by measuring vehicle exhaust emissions at the request of JATA (Japan Automobile Transport Technology Association). Since the data is for a standard car using 3,000-liter unleaded premium gasoline, further reductions can be expected for diesel and heavy oil.

8 value ppm NDIR
What is CO value?
When the supply of oxygen is insufficient during combustion, incomplete combustion occurs, resulting in harmful emission gases (Carbon monoxide)

30.4%D0WN



ual check revealed a clear difference confirming that the exhaust gas was

HC value ppmC FID What is **HC** value?

This value is the **unburned** portion of the fuel. The improved combustion efficiency in the engine results in lower **HC** values.

23.3%D0WN

NOx value ppm CLD **What is NOx** value?

Nitrogen oxides are produced when combustion conditions are good or efficient.

" Reduced even in good combustion conditions by using **eco-spray**) **15.09%D0WN**

CO_2 reduction was 0.43% in the above experiment

Confidential 14

Appendix 1

Reiwa **August 25 , 5**

Customer Name	Hiroshima Commercial Technical College	inspector	Kyowa **1: Kogyo** K.K. Wada
Effective Date	concord **5** years **August twenty-fifth** day of the month		
use na	light		
meengine type	Komatsu **6M10K4PT**	Institution No.	Starboard #21813
operating time question	**1875 Hr**		Left glare #21814

Construc tion name	
	stem open maintenance
4: Business	

1) **Supercharger** open inspection
2) **Cylinder head** open inspection
3) **Intake manifold** open inspection
4) **Exhaust manifold** open inspection
5) Offshore **iit** operation
remarks
There was no significant difference in the internal parts of the engine and the intake and exhaust systems on both sides of the vessel during this maintenance.
No corrosion or other specific effects due to chemical spraying on the port plane were observed.
Both port and starboard planes were in good condition.

August 25 , **2023**

Notice of Completion of Construction

Javan Family SabotCo.

Inspection work for the training ship "Hikari"

Period during which construction was performed

Indication of construction

August 18 , 2023 August 25 , 2023

open for joint research

We hereby notify you that the above has been completed.

contractor

32 Takara, Incho , Mimocho, Onomichi City, Hiroshima Prefecture 9 3. 1-4

Appendix 2

Reiwa December 1 , 5 Javan

Family Support Co., Ltd. * Research Representative (Affiliation - - Name - Name)

Yorimilwahata, Assistant Professor, Department of Merchant Marine

RERF

We are pleased to announce that the joint research contracted on September 1 , 2002 has been completed as follows : I will. (1) Research Title : period

Research on the effectiveness of eco-SPRAX8 on ships (2) Summary of research results: In the second study, we conducted a demonstration test of eco-spray, a fuel additive for automobiles ,for use in marine vessels, and obtained information on the effects ofeco-spray on the main engine. The school ' s training ship , Hikari, was used . As in the case of the demonstration for the effect - must, we conducted a total of86 navigation experiments (about1300[m]) over the course of one year, using Isets of40-minute cruises (20 minutes for the outwardcruise and 2 minutes for the return cruise, for a total of 40 minutes, in order totake into consideration the external disturbance of the national wave specialist), andcalculated the total consumption of fees at . After the actual run , an inspection was also conducted on the main engine and the final condition of the engine was confirmed.

Results are available at:**Fig. As** shown in Fig. **I,** the low fuel consumption rate of 0.48% was eliminated. As shown in **2** A change in the color **OF** the exhaust from the scavenging pipe was **observed** , and it was confirmed that white smoke was suppressed from the first test **USING**

Based on these results, the demonstration test **of** eco-SPRAY ♦ for ship supply was completed without any problems . The effectiveness **of** using eco-SPRAY on ships was confirmed.

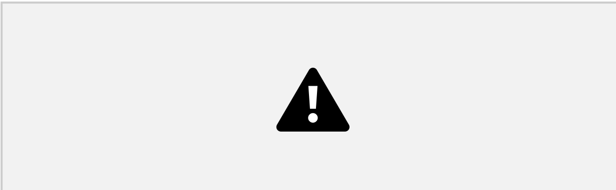
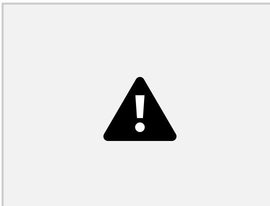
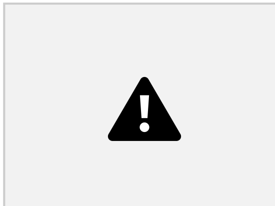


Fig. Change in fuel cost by changing the analysis Fig. 2

Comparison of stock gases for ships and boats (for 10-ship transport, for eco-S«AY transport)



Fi" 3 M.Silga F. Doda at the time of drumming, "unused "



Fit4 m shell i¢--6si, "f dai "eco-SraZ used)

SIspray . **The main engine** manufacturer conducted an overhaul inspection, which revealed that no abnormalities were found in the intake and exhaust systems . The combustion conditions were good. **As** shown in **Figures** 3 and 4 , changes in the amount of carbon attached to the

別紙1

別紙2